Title: Date: / / M T W T F S S

Quiz 2

Hard Mode

$\ln(x) = 2$ using 5 iterations
$x_0 = 2$

$\ln(x) = 2$

$\downarrow$

$f(x) = \ln(x) - 2$

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

$f'(x) = 1/2$ $\rightarrow$ $x_{n+1} = x_n - \frac{\ln(x_n) - 2}{1/x_n} = x_n - x_n[\ln(x_n) - 2]$

$= x_{n+1} = x_n[3 - \ln(x_n)]$

$x_0 = 2$

it: 1

$n=0$

$x_0 = 2$

$\ln(2) = 0.6931$

$3 - \ln(2) = 2.3068$

$x_1 = 2 \times 2.3068 = 4.6137.$

iteration 2:

$x_1 \rightarrow 4.6137$

$\ln(4.6137) \approx 1.5285$

$3 - 1.5285 = 1.4715$

$x_2 = 4.6137 \times 1.4715 = 6.7879$

it: 3

$x_2 = 6.7879$

$\ln(6.7879) \approx 1.9145$

$3 - 1.9145 \approx 1.085$

$x_3 = 6.7879 \times 1.085 \approx 7.368$

iter: 4

$x_3 = 7.368$

$\ln(7.368) \approx 1.997$

$3 - 1.9971 = 1.0028$

$x_4 = 7.368 \times 1.0028 \approx 7.388$

Exact: $x = e^2 = 7.389$

iter: 5

$x_4 = 7.388$

$\ln(7.388) \approx 2.000021$

$3 - 2.000021 = 0.9999$

$x_5 = 7.388 \times 0.9999 = 7.389$

$x_5 = 7.389$